

Suyash Naresh Madhavi

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Data Engineer with hands-on experience building cloud-native ETL/ELT pipelines, data workflows on AWS, GCP, and Azure. Skilled in Python, SQL, Spark, and Airflow, with a track record of automating data systems and deploying production-grade pipelines.

Experience

Backend Developer Intern

Feb 2023 – June 2023

GrayQuest

- Automated database workflows by creating Python scripts and SQL queries, reducing manual processing time.
- Developed REST APIs using Django for data logs, improving system monitoring.
- Conducted production webpage testing using Jira, identifying and resolving critical issues.

CRM Workflow & Analytics Volunteer

July 2026 – Present

Hebron Community Center

- Debugged and managed Zoho Creator and CRM workflows, streamlining backend operations and resolving critical logic errors.
- Designed and deployed interactive analytics dashboards to visualize operational data and support strategic decision-making.
- Engineered automated check-in systems via custom Deluge scripts and instituted a structured code deployment lifecycle across development and staging environments.

Education

Stevens Institute of Technology

Hoboken, NJ

Master of Science in Computer Science

May 2026

University of Mumbai

Mumbai, MH

Bachelor of Engineering in Computer Engineering

August 2022

Skills

Languages / Databases: Python, PySpark, C, SQL, Java, MySQL, PostgreSQL, GraphQL, Data Lake/Warehouse, Pinecone

Frameworks / Tools: TensorFlow, Power BI, Docker, GCP, Apache Airflow, Kafka, Flink, AWS, Git, Azure, FastAPI, dbt, Snowflake, Zoho Analytics, ETL/ELT, Databricks

Academic Projects

Real-Time F1 Telemetry Analytics Pipeline

June 2026

- Built a real-time F1 telemetry pipeline using Kafka and PySpark Structured Streaming across six Kafka partitions.
- Integrated checkpointed micro-batch Snowflake writes using foreachBatch and idempotent MERGE; validated with 150,000 events at 1,000 events/second.
- Calculated an explainable theoretical optimal lap by summing the best observed corner segments across drivers and laps.

FAIR Oncology Omics Data Pipeline & Metadata Platform

August 2026

- Architected an ETL pipeline using Airflow, PySpark, Podman to continuously ingest, clean, and standardize oncology datasets.
- Designed normalized PostgreSQL database schemas using SQL, R data wrangling scripts, to resolve missing values.
- Integrated a LangChain agent to automate metadata extraction, validate identifier, optimize dataset quality control workflows.

Context-Aware AI Routing System

Jan 2026

- Built a Federated RAG application using FastAPI and Gemini for intelligent routing.
- Orchestrated AWS data pipelines with Apache Airflow for document extraction and embeddings.
- Integrated Pinecone with PostgreSQL for multi-source retrieval and ranking.

Serverless AI Web App on AWS

Apr 2026

- Developed a secure serverless web application using S3, Lambda, and CloudFront.
- Built a Python-based sentiment analysis and Flesch-Kincaid readability scoring algorithm.
- Secured the application with HTTPS via CloudFront and monitored backend services using CloudWatch.

FMCG Lakehouse Merger ETL Pipeline with Databricks

Dec 2025

- Engineered a Databricks ETL pipeline integrating datasets from two FMCG companies into a unified lakehouse.
- Built scalable workflows using Python, SQL, Spark, AWS S3, and the Medallion architecture.
- Delivered BI dashboards on curated and serving layers for enterprise analytics.

Diabetic Retinopathy Severity Classification

May 2025

- Built an end-to-end image classification pipeline using PySpark for ETL and PyTorch CNN models.
- Processed retinal fundus images and trained a multi-class classification model.

Certifications & Awards

- Stevens QuackHack 2026 Winner
- AWS Certified Data Engineer – Associate
- Astronomer Apache Airflow 3 Fundamentals